Y3S2 Group Project Explaining the Code

jxm1517

February 2026

1 Radiative Transfer

When forming a model absorption spectrum, our code takes the input planetary parameters and assumes a one-dimensional, isothermal atmosphere with a constant composition ratio, where pressure decreases with radial distance according to $p(r) \propto \frac{1}{r^2}$. The atmosphere is assumed to behave as an ideal gas, so the number density is calculated as $n = \frac{p}{kT}$. The cloud deck (at a specified pressure) is treated as a hard cutoff and is assumed to be 100% opaque below that level.

The opacity is calculated wavelength-by-wavelength from the cross-section files for each gas. These values are interpolated and then weighted by number density and mixing ratio before being combined to calculate the transit depth at each wavelength.